**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Lourenco/Wickstrom Dairy #4

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 16126 Sunset AVE | Livingston | Merced | 95334 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/23/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X049-X030-X038-XXXX

**B. OPERATORS**

Wickstrom, Aaron

Operator name: Wickstrom, Aaron

Telephone no.: (209) 667-7055 Landline Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| P.O. Box 880 | Hilmar | CA | 95324 |

**This operator is responsible for paying permit fees.**

Wickstrom, Tim

Operator name: Wickstrom, Tim

Telephone no.: (209) 667-7055 Landline Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| P.O. Box 880 | Hilmar | CA | 95324 |

**C. OWNERS**

Lourenco, John

Legal owner name: Lourenco, John

Telephone no.: (209) 667-7055 Landline Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 7170 S Robin AVE | Livingston | CA | 95334 |

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 0 | 0 | 0 | 0 |
| Number under roof | 0 | 140 | 108 | 0 | 0 | 0 |
| Maximum number | 0 | 280 | 216 | 0 | 0 | 0 |
| Average number | 0 | 140 | 108 | 0 | 0 | 0 |
| Avg live weight (lbs) | 0 | 1,200 | 700 | 0 | | |

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 2,927.50 *tons per reporting period*

Total nitrogen from manure: 35,799.20 *lbs per reporting period*

After ammonia losses (30% loss applied): 25,059.44 *lbs per reporting period*

Total phosphorus from manure: 5,107.08 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 32,193.00 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

| | |
|---|---|
| | 0 *gallons applied* |
| + | 0 *gallons exported* |
| - | 0 *gallons imported* |
| = | 0 *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Canal | Surface water |
| Irrigation Well | Ground water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

| Date | Material type / Description | Quantity | Reporting basis | Moisture (%) | N (%) | P (%) | K (%) | Salt (%) |
|---|---|---|---|---|---|---|---|---|
| 04/01/2023 | Solid commercial fertilizer<br>*NH3* | 8.63 *ton* | Dry-weight | 0.1 | 82.000000 | 0.000000 | 0.000000 | 0.000000 |
| 07/16/2023 | Solid commercial fertilizer<br>*UN32* | 7.00 *ton* | Dry-weight | 0.1 | 32.000000 | 0.000000 | 0.000000 | 0.000000 |

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Commercial fertilizer / Other | 18,614.57 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total imports for all materials | 18,614.57 | 0.00 | 0.00 | 0.00 |

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| 1245 Pivot | 150 | 150 | 2 | none | X049-X030-X038-XXXX |
| 3 | 37 | 37 | 2 | none | X049-X040-X014-XXXX |
| 3a | 15 | 15 | 2 | none | X049-X040-X014-XXXX |
| 6789 Pivot | 140 | 140 | 2 | none | X049-X040-X011-XXXX |
| Totals for areas that were used for application | | | | | |
| Totals for areas that were not used for application | 342 | 342 | 8 | | |
| Land application area totals | 342 | 342 | 8 | | |

## B. CROPS AND HARVESTS

1245 Pivot

Field name: 1245 Pivot

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 150 Plant date: 11/14/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/25/2023 | 1,807.98 *ton* | As-is | | 53.6 | 4,700.00 | 900.00 | 3,700.00 | | 6.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 12.05 | 113.30 | 21.70 | 89.19 | 749.42 |

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 150 Plant date: 06/09/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/30/2023 | 3,992.75 *ton* | As-is | | 67.0 | 4,100.00 | 400.00 | 1,500.00 | | 3.79 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.62 | 218.27 | 21.29 | 79.86 | 665.83 |

3

Field name: 3

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 37 — Plant date: 11/14/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/25/2023 | 425.23 *ton* | As-is | | 53.8 | 6,500.00 | 900.00 | 4,300.00 | | 6.05 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 11.49 | 149.41 | 20.69 | 98.84 | 642.47 |

06/09/2023: Corn, silage

Crop: Corn, silage — Acres planted: 37 — Plant date: 06/09/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/30/2023 | 999.46 *ton* | As-is | | 71.8 | 3,800.00 | 400.00 | 3,700.00 | | 5.41 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 27.01 | 205.29 | 21.61 | 199.89 | 824.21 |

3a

Field name: 3a

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 15 — Plant date: 11/15/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/25/2023 | 102.23 *ton* | As-is | | 40.8 | 9,000.00 | 1,400.00 | 16,100.00 | | 9.64 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 6.82 | 122.68 | 19.08 | 219.45 | 777.89 |

3a

06/09/2023: Corn, silage

Crop: Corn, silage — Acres planted: 15 — Plant date: 06/09/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/30/2023 | 448.69 *ton* | As-is | | 68.2 | 3,700.00 | 400.00 | 3,400.00 | | 4.49 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 29.91 | 221.35 | 23.93 | 203.41 | 854.20 |

6789 Pivot

Field name: 6789 Pivot

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough — Acres planted: 140 — Plant date: 11/15/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/24/2023 | 1,511.54 *ton* | As-is | | 51.4 | 6,100.00 | 900.00 | 4,300.00 | | 6.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 10.80 | 131.72 | 19.43 | 92.85 | 629.66 |

06/10/2023: Corn, silage

Crop: Corn, silage — Acres planted: 140 — Plant date: 06/10/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/29/2023 | 3,773.26 *ton* | As-is | | 67.0 | 4,000.00 | 500.00 | 1,800.00 | | 4.96 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 26.95 | 215.61 | 26.95 | 97.03 | 882.30 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### 1245 Pivot - 11/14/2022: Oats, silage-soft dough

Field name: 1245 Pivot

Crop: Oats, silage-soft dough — Plant date: 11/14/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/01/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 55.93 | 0.00 | 0.00 | 770.00 | 24,281,700.00 *gal* |
| Application event totals | | 55.93 | 0.00 | 0.00 | 770.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Solid commercial fertilizer | 41.00 | 0.00 | 0.00 | 0.00 | |
| Irrigation Well | Ground water | 45.78 | 0.00 | 0.00 | 630.33 | 19,877,250.00 *gal* |
| Application event totals | | 86.78 | 0.00 | 0.00 | 630.33 | |

### 1245 Pivot - 06/09/2023: Corn, silage

Field name: 1245 Pivot

Crop: Corn, silage — Plant date: 06/09/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 88.14 | 0.00 | 0.00 | 1,213.53 | 38,268,550.00 *gal* |
| Application event totals | | 88.14 | 0.00 | 0.00 | 1,213.53 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 69.99 | 0.00 | 0.00 | 963.69 | 30,389,700.00 *gal* |
| Application event totals | | 69.99 | 0.00 | 0.00 | 963.69 | |

1245 Pivot - 06/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 64.60 | 0.00 | 0.00 | 889.44 | 28,048,200.00 *gal* |
| Application event totals | | 64.60 | 0.00 | 0.00 | 889.44 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 73.95 | 0.00 | 0.00 | 1,018.14 | 32,106,721.00 *gal* |
| Application event totals | | 73.95 | 0.00 | 0.00 | 1,018.14 | |

3 - 11/14/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough Plant date: 11/14/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/01/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 53.47 | 0.00 | 0.00 | 841.62 | 6,271,574.00 *gal* |
| Application event totals | | 53.47 | 0.00 | 0.00 | 841.62 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Solid commercial fertilizer | 41.00 | 0.00 | 0.00 | 0.00 | |
| Irrigation Well | Ground water | 42.04 | 0.00 | 0.00 | 661.70 | 4,930,842.00 *gal* |
| Application event totals | | 83.04 | 0.00 | 0.00 | 661.70 | |

3 - 06/09/2023: Corn, silage

Field name: 3

Crop: Corn, silage Plant date: 06/09/2023

3 - 06/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 77.58 | 0.00 | 0.00 | 1,221.11 | 9,099,435.00 *gal* |
| Application event totals | | 77.58 | 0.00 | 0.00 | 1,221.11 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 69.45 | 0.00 | 0.00 | 1,093.13 | 8,145,735.00 *gal* |
| Application event totals | | 69.45 | 0.00 | 0.00 | 1,093.13 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 78.49 | 0.00 | 0.00 | 1,235.44 | 9,206,175.00 *gal* |
| Application event totals | | 78.49 | 0.00 | 0.00 | 1,235.44 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 72.20 | 0.00 | 0.00 | 1,136.53 | 8,469,150.00 *gal* |
| Application event totals | | 72.20 | 0.00 | 0.00 | 1,136.53 | |

3a - 11/15/2022: Oats, silage-soft dough

Field name: 3a

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/28/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 50.20 | 2,438,925.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 50.20 | |

3a - 11/15/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Solid commercial fertilizer | 123.00 | 0.00 | 0.00 | 0.00 | |
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.76 | 1,980,375.00 *gal* |
| Application event totals | | 123.00 | 0.00 | 0.00 | 40.76 | |

3a - 06/09/2023: Corn, silage

Field name: 3a

Crop: Corn, silage

Plant date: 06/09/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 48.13 | 2,338,125.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 48.13 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.78 | 1,738,125.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.78 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/18/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UN32 | Solid commercial fertilizer | 96.00 | 0.00 | 0.00 | 0.00 | |
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 34.96 | 1,698,240.00 *gal* |
| Application event totals | | 96.00 | 0.00 | 0.00 | 34.96 | |

3a - 06/09/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UN32 | Solid commercial fertilizer | 64.00 | 0.00 | 0.00 | 0.00 | |
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 32.18 | 1,563,240.00 *gal* |
| Application event totals | | 64.00 | 0.00 | 0.00 | 32.18 | |

| | | | | |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| UN32 | Solid commercial fertilizer | 64.00 | 0.00 | 0.00 | 0.00 | |
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 29.13 | 1,415,120.00 *gal* |
| Application event totals | | 64.00 | 0.00 | 0.00 | 29.13 | |

| | | | | |
|---|---|---|---|---|
| 08/15/2023 | Pipeline | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 27.74 | 1,347,725.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 27.74 | |

| | | | | |
|---|---|---|---|---|
| 08/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 22.81 | 1,108,113.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 22.81 | |

| | | | | |
|---|---|---|---|---|
| 09/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 20.37 | 989,423.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 20.37 | |

| | | | | |
|---|---|---|---|---|
| 09/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Canal | Surface water | 0.00 | 0.00 | 0.00 | 16.91 | 821,560.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 16.91 | |

6789 Pivot - 11/15/2022: Oats, silage-soft dough

Field name: 6789 Pivot

Crop: Oats, silage-soft dough — Plant date: 11/15/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/01/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 55.80 | 0.00 | 0.00 | 436.39 | 24,003,700.00 *gal* |
| Application event totals | | 55.80 | 0.00 | 0.00 | 436.39 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| NH3 | Solid commercial fertilizer | 41.00 | 0.00 | 0.00 | 0.00 | |
| Irrigation Well | Ground water | 45.81 | 0.00 | 0.00 | 358.24 | 19,705,000.00 *gal* |
| Application event totals | | 86.81 | 0.00 | 0.00 | 358.24 | |

6789 Pivot - 06/10/2023: Corn, silage

Field name: 6789 Pivot

Crop: Corn, silage — Plant date: 06/10/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 84.11 | 0.00 | 0.00 | 657.80 | 36,182,196.00 *gal* |
| Application event totals | | 84.11 | 0.00 | 0.00 | 657.80 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 69.46 | 0.00 | 0.00 | 543.18 | 29,877,823.00 *gal* |
| Application event totals | | 69.46 | 0.00 | 0.00 | 543.18 | |

6789 Pivot - 06/10/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 72.22 | 0.00 | 0.00 | 564.80 | 31,067,024.00 *gal* |
| Application event totals | | 72.22 | 0.00 | 0.00 | 564.80 | |

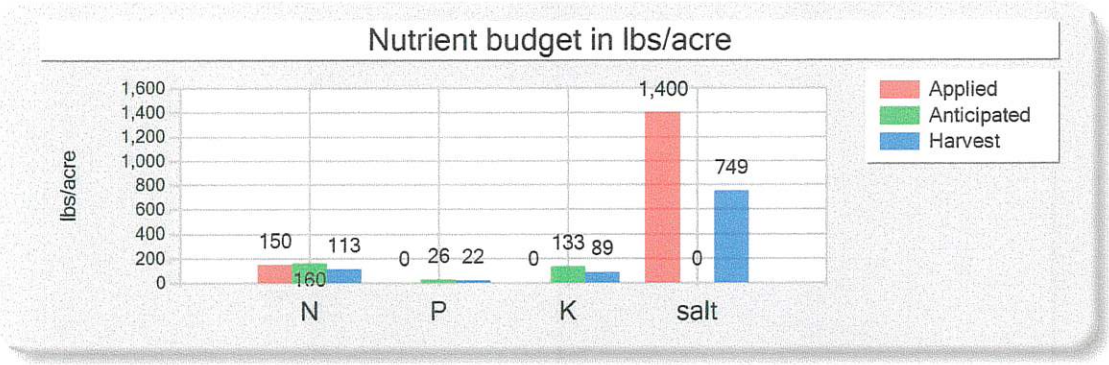
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Well | Ground water | 61.42 | 0.00 | 0.00 | 480.35 | 26,421,621.00 *gal* |
| Application event totals | | 61.42 | 0.00 | 0.00 | 480.35 | |

## B. NUTRIENT BUDGET

1245 Pivot - 11/14/2022: Oats, silage-soft dough

Field name: 1245 Pivot  Crop: Oats, silage-soft dough  Plant date: 11/14/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 41.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 101.71 | 0.00 | 0.00 | 1,400.32 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 149.71 | 0.00 | 0.00 | 1,400.32 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 113.30 | 21.70 | 89.19 | 749.42 |
| Nutrient balance | 36.41 | -21.70 | -89.19 | 650.90 |
| Applied to removed ratio | 1.32 | 0.00 | 0.00 | 1.87 |

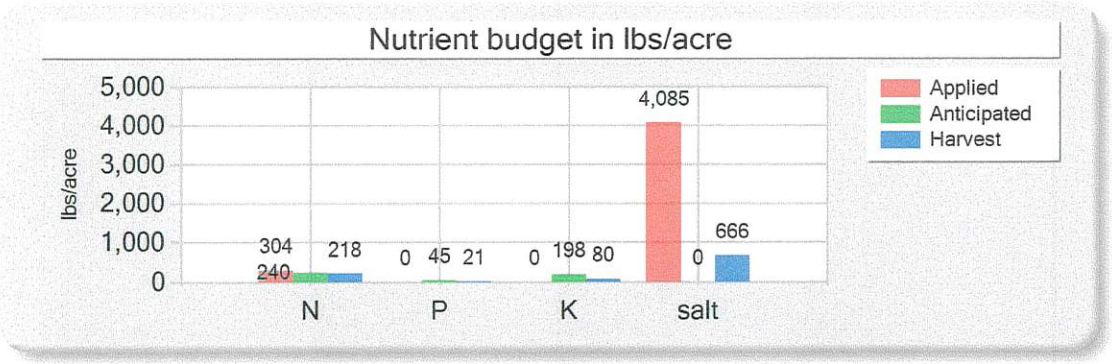
| Fresh water applied |
|---|
| 44,158,950.00 *gallons* |
| 1,626.22 *acre-inches* |
| 10.84 *inches/acre* |

| Process wastewater applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

1245 Pivot - 06/09/2023: Corn, silage

Field name: 1245 Pivot  Crop: Corn, silage  Plant date: 06/09/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 296.69 | 0.00 | 0.00 | 4,084.79 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 303.69 | 0.00 | 0.00 | 4,084.79 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 218.27 | 21.29 | 79.86 | 665.83 |
| Nutrient balance | 85.41 | -21.29 | -79.86 | 3,418.96 |
| Applied to removed ratio | 1.39 | 0.00 | 0.00 | 6.13 |

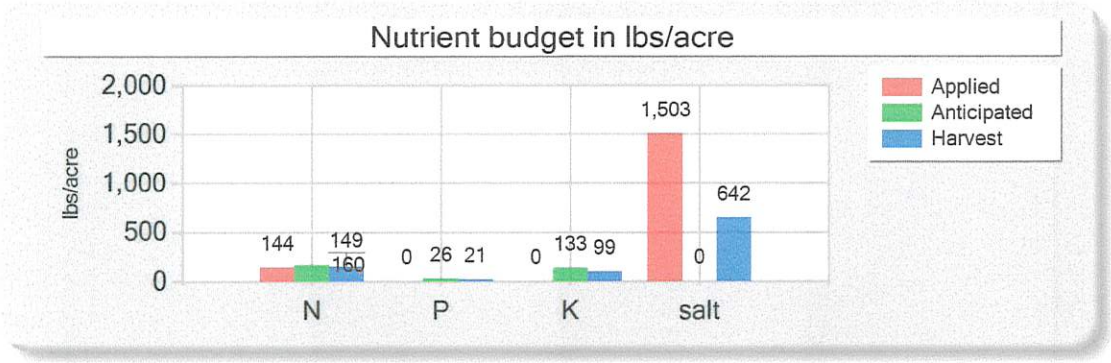
| Fresh water applied |
|---|
| 128,813,171.00 *gallons* |
| 4,743.75 *acre-inches* |
| 31.63 *inches/acre* |

| Process wastewater applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

3 - 11/14/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/14/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 41.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 95.51 | 0.00 | 0.00 | 1,503.33 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 143.51 | 0.00 | 0.00 | 1,503.33 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 149.41 | 20.69 | 98.84 | 642.47 |
| Nutrient balance | -5.90 | -20.69 | -98.84 | 860.86 |
| Applied to removed ratio | 0.96 | 0.00 | 0.00 | 2.34 |

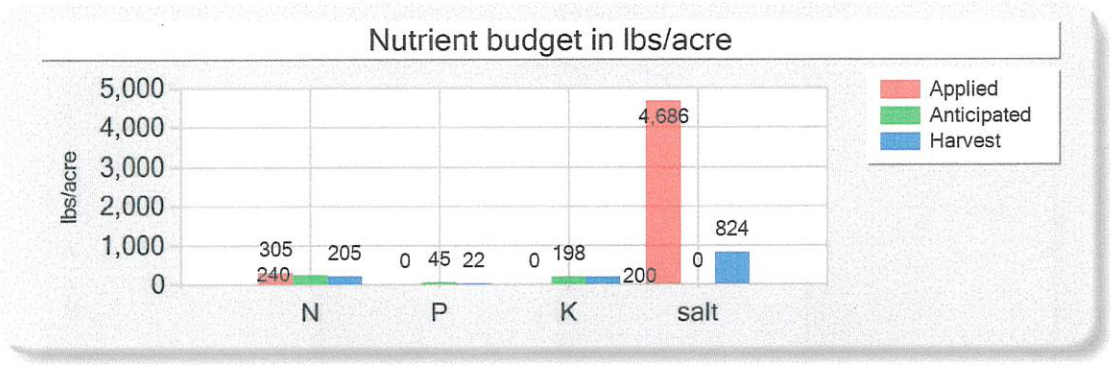
| Fresh water applied |
|---|
| 11,202,416.00 *gallons* |
| 412.55 *acre-inches* |
| 11.15 *inches/acre* |

| Process wastewater applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

3 - 06/09/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/09/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 297.71 | 0.00 | 0.00 | 4,686.21 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 304.71 | 0.00 | 0.00 | 4,686.21 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 205.29 | 21.61 | 199.89 | 824.21 |
| Nutrient balance | 99.42 | -21.61 | -199.89 | 3,862.00 |
| Applied to removed ratio | 1.48 | 0.00 | 0.00 | 5.69 |

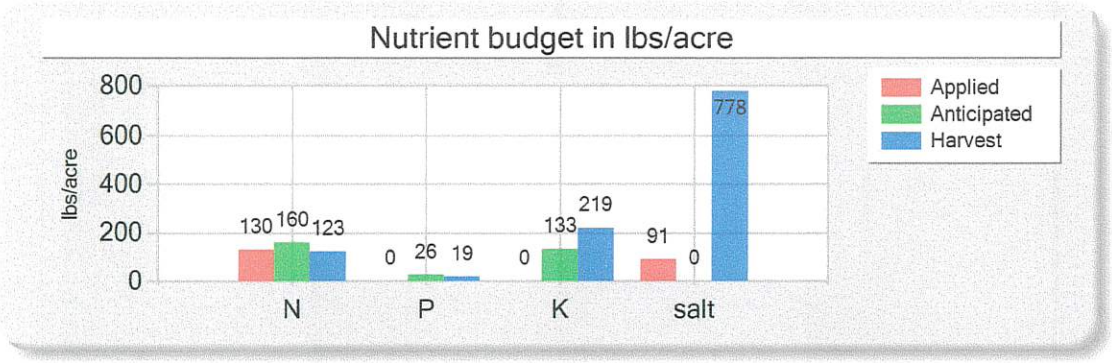
| Fresh water applied | |
|---|---|
| 34,920,495.00 | *gallons* |
| 1,286.00 | *acre-inches* |
| 34.76 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

3a - 11/15/2022: Oats, silage-soft dough

Field name: 3a Crop: Oats, silage-soft dough Plant date: 11/15/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 123.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 0.00 | 0.00 | 0.00 | 90.97 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 130.00 | 0.00 | 0.00 | 90.97 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 122.68 | 19.08 | 219.45 | 777.89 |
| Nutrient balance | 7.32 | -19.08 | -219.45 | -686.92 |
| Applied to removed ratio | 1.06 | 0.00 | 0.00 | 0.12 |

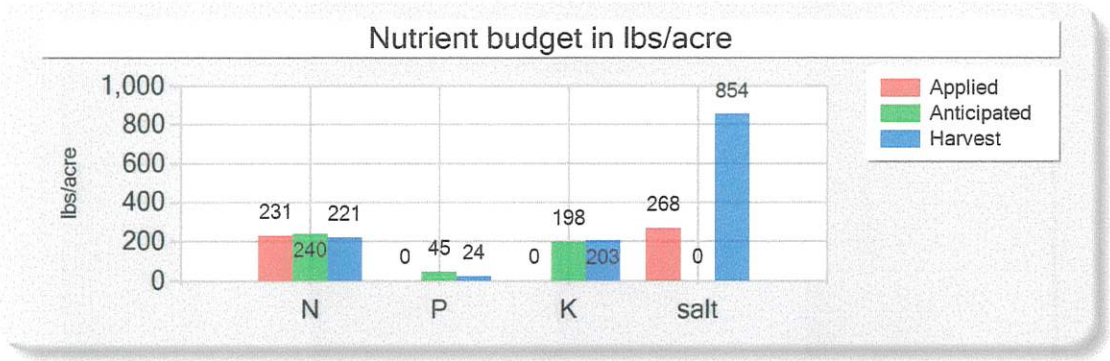
| Fresh water applied |
|---|
| 4,419,300.00 *gallons* |
| 162.75 *acre-inches* |
| 10.85 *inches/acre* |

| Process wastewater applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

3a - 06/09/2023: Corn, silage

Field name: 3a  Crop: Corn, silage  Plant date: 06/09/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 224.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 0.00 | 0.00 | 0.00 | 268.00 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 231.00 | 0.00 | 0.00 | 268.00 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 221.35 | 23.93 | 203.41 | 854.20 |
| Nutrient balance | 9.65 | -23.93 | -203.41 | -586.20 |
| Applied to removed ratio | 1.04 | 0.00 | 0.00 | 0.31 |

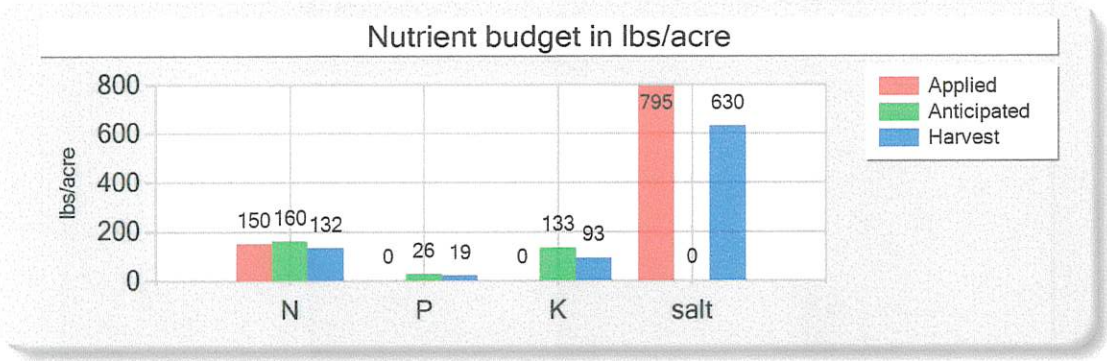
| Fresh water applied |
|---|
| 13,019,671.00 *gallons* |
| 479.47 *acre-inches* |
| 31.96 *inches/acre* |

| Process wastewater applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

6789 Pivot - 11/15/2022: Oats, silage-soft dough

Field name: 6789 Pivot  Crop: Oats, silage-soft dough  Plant date: 11/15/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 41.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 101.61 | 0.00 | 0.00 | 794.63 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 149.61 | 0.00 | 0.00 | 794.63 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 131.72 | 19.43 | 92.85 | 629.66 |
| Nutrient balance | 17.89 | -19.43 | -92.85 | 164.97 |
| Applied to removed ratio | 1.14 | 0.00 | 0.00 | 1.26 |

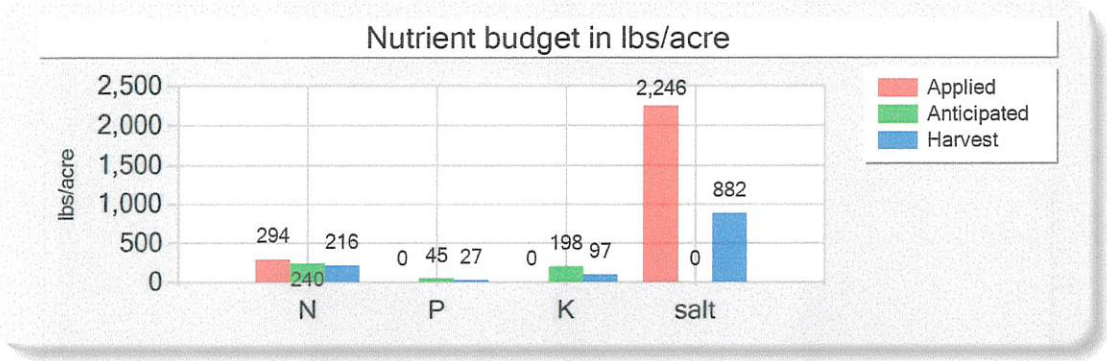
| Fresh water applied |
|---|
| 43,708,700.00 *gallons* |
| 1,609.64 *acre-inches* |
| 11.50 *inches/acre* |

| Process wastewater applied |
|---|
| 0.00 *gallons* |
| 0.00 *acre-inches* |
| 0.00 *inches/acre* |

| Total harvests for the crop |
|---|
| 1 *harvests* |

6789 Pivot - 06/10/2023: Corn, silage

Field name: 6789 Pivot  Crop: Corn, silage  Plant date: 06/10/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 287.21 | 0.00 | 0.00 | 2,246.14 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 294.21 | 0.00 | 0.00 | 2,246.14 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 215.61 | 26.95 | 97.03 | 882.30 |
| Nutrient balance | 78.60 | -26.95 | -97.03 | 1,363.84 |
| Applied to removed ratio | 1.36 | 0.00 | 0.00 | 2.55 |

| Fresh water applied | |
|---|---|
| 123,548,664.00 | *gallons* |
| 4,549.88 | *acre-inches* |
| 32.50 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 11,100.00 | 1,600.00 | 3,500.00 | | | | | | | 20.10 |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | 0.01 |

### Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 8,400.00 | 6,100.00 | 21,800.00 | 14,100.00 | 5,600.00 | 1,500.00 | 2,600.00 | 891.00 | | 73.00 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | | 0.01 |

## B. PROCESS WASTEWATER ANALYSES

### WWSQ1

Sample and source description: WWSQ1

Sample date: 03/29/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 300.00 | 186.00 | 0.00 | 0.00 | 93.10 | 812.00 | | | | | | | | 4,540.00 | 4,740 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 269.00 | 129.00 | 0.00 | 0.00 | 56.80 | 925.00 | | | | | | | | 4,480.00 | 3,400 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 165.00 | 109.00 | 0.00 | 0.00 | 26.10 | 221.00 | 51.30 | 28.50 | 117.00 | 1,020.00 | 0.00 | 31.80 | 212.00 | 2,860.00 | 1,660 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 1.00 | 10 |

WWSQ4

Sample and source description: WWSQ4

Sample date: 10/05/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: _____

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 230.00 | 137.00 | 0.00 | 0.00 | 42.30 | 363.00 | | | | | | | | 3,500.00 | 1,660 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## C. FRESH WATER ANALYSES

Canal

## Canal

### MID Canal

Sample description: MID Canal

Sample date: 07/25/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | 0.00 | 0.00 | | | | | | | | 38.10 | 37 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## Irrigation Well

### IW #1

Sample description: IW #1

Sample date: 12/19/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 37.80 | 0.00 | 37.80 | | | | | | | | 1,010.00 | 590 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

### IW #2

Sample description: IW #2

Sample date: 12/19/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 41.40 | 0.00 | 38.60 | | | | | | | | 1,000.00 | 570 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

Irrigation Well

IW #3

Sample description: IW #3

Sample date: 12/19/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 37.80 | 0.00 | 37.80 | | | | | | | | 1,010.00 | 595 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

IW #4

Sample description: IW #4

Sample date: 12/19/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 38.30 | 0.00 | 38.30 | | | | | | | | 1,020.00 | 615 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

IW #5

Sample description: IW #5

Sample date: 12/19/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 39.00 | 0.00 | 38.80 | | | | | | | | 1,000.00 | 305 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

1245 Pivot - 11/14/2022: Oats, silage-soft dough

1245 Pivot - 11/14/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,300.00 | 800.00 | 3,500.00 | | 5.62 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Winter Forage

Sample and source description: Winter Forage
Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 53.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,700.00 | 900.00 | 3,700.00 | | 6.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Winter Forage

Sample and source description: Winter Forage
Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,400.00 | 600.00 | 3,300.00 | | 6.48 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

1245 Pivot - 11/14/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,800.00 | 800.00 | 3,500.00 | | 6.38 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

1245 Pivot - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,100.00 | 400.00 | 1,500.00 | | 3.79 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Corn Silage

Sample and source description: Corn Silage
Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 75.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 2,900.00 | 400.00 | 1,900.00 | | 5.10 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

1245 Pivot - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,500.00 | 500.00 | 1,400.00 | | 4.47 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

3 - 11/14/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 6,500.00 | 900.00 | 4,300.00 | | 6.05 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

3 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,800.00 | 400.00 | 3,700.00 | | 5.41 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

3a - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 40.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,000.00 | 1,400.00 | 16,100.00 | | 9.64 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

3a - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,700.00 | 400.00 | 3,400.00 | | 4.49 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

6789 Pivot - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,700.00 | 900.00 | 4,600.00 | | 6.75 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

6789 Pivot - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 6,100.00 | 900.00 | 4,300.00 | | 6.00 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Winter Forage

Sample and source description: Winter Forage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,200.00 | 400.00 | 2,400.00 | | 6.39 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

Winter Forage

Sample and source description: Winter Forage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 3,800.00 | 700.00 | 3,800.00 | | 7.25 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

6789 Pivot - 06/10/2023: Corn, silage

6789 Pivot - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 4,000.00 | 500.00 | 1,800.00 | | 4.96 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

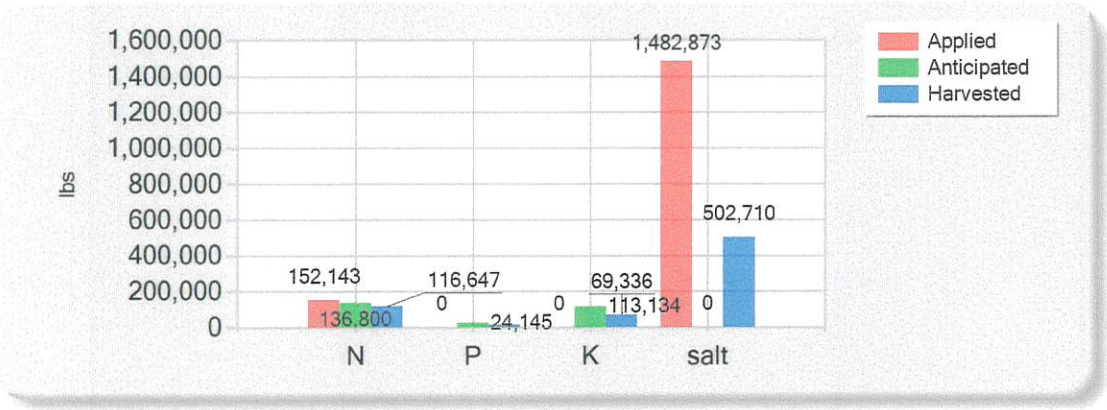
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

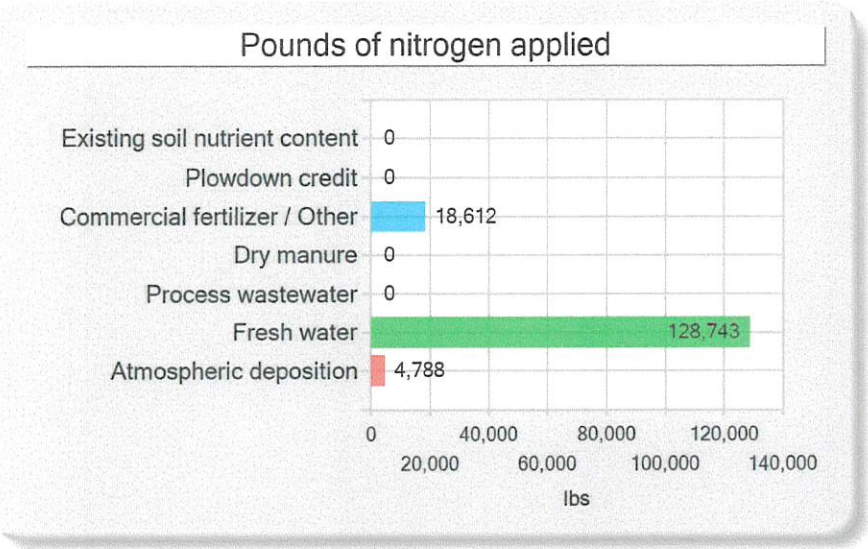
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

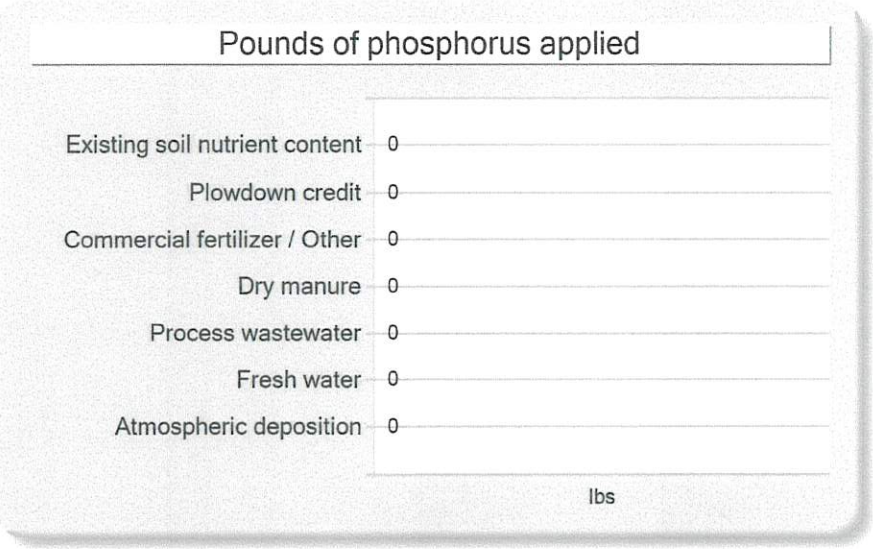
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 18,612.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 128,742.73 | 0.00 | 0.00 | 1,482,872.94 |
| Atmospheric deposition | 4,788.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 152,142.73 | 0.00 | 0.00 | 1,482,872.94 |
| Anticipated crop nutrient removal | 136,800.00 | 24,145.20 | 113,133.60 | 0.00 |
| Actual crop nutrient removal | 116,646.76 | 15,152.77 | 69,336.16 | 502,710.45 |
| Nutrient balance | 35,495.97 | -15,152.77 | -69,336.16 | 980,162.49 |
| Applied to removed ratio | 1.30 | 0.00 | 0.00 | 2.95 |

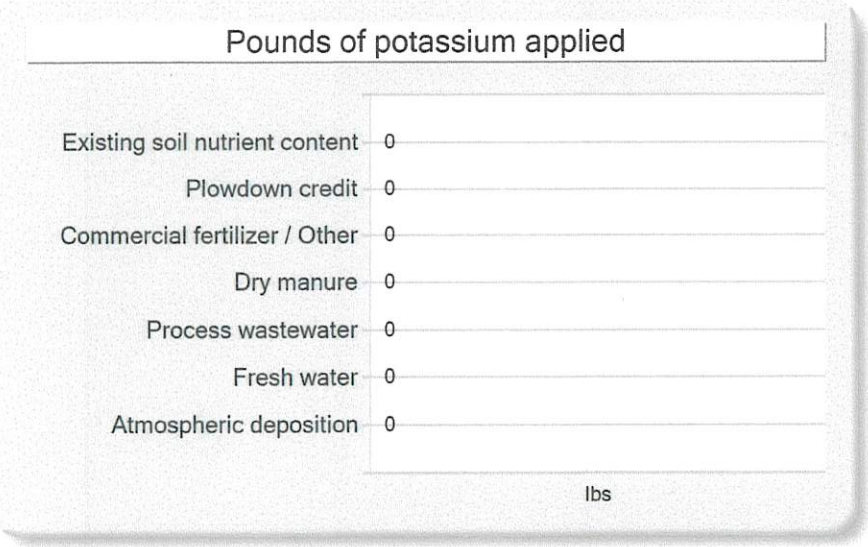
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

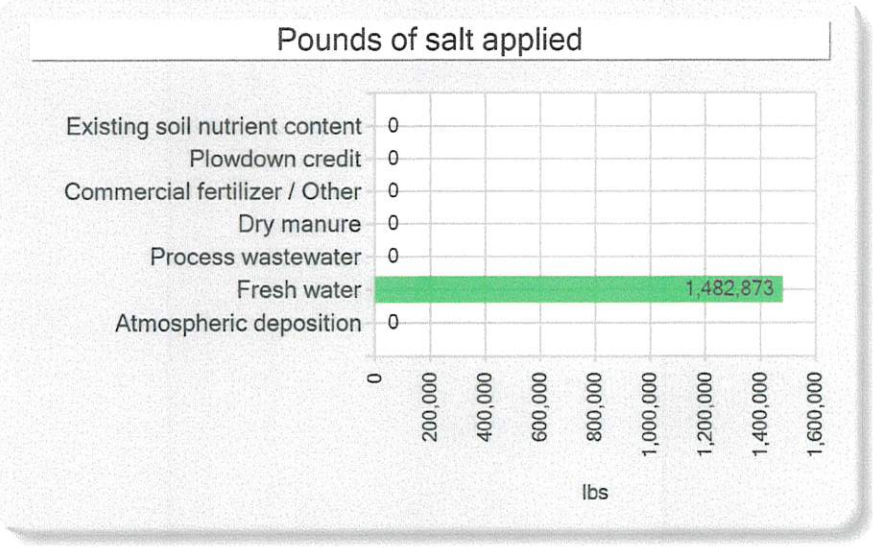


**C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE**



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | No |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

## ADDITIONAL NOTES

### A. NOTES

Facility only housed dry cows and bred heifers until June of 2023. Facility remained empty for the rest of the year.

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| On behalf of John Lourenco | |
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| John Lourenco | Aaron Wickstrom |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 6/6/2024 | 6/6/2024 |
| DATE | DATE |

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed. |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

# DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

**Client:** Wickstrom/Lourenco
**Mailing address:**
(not facility address)
**Phone#:**

**Authorized Copy Released To:** F&R AG SERVICES
**Sampled By:**
**Project ID:** K.Jones

## ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

**TILE DRAIN**
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

**PLANT TISSUE**
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 12/19 | 7:00 | Domestic Wells | DW 6 | 14.8°c | | X | | | | |
| 2 | | 7:03 | Domestic Wells | DW 7 | 14.6°c | | X | | | | |
| 3 | | 7:08 | Irrigation Wells | IW 1 | 15.8°c | | | | | | H7 |
| 4 | | 7:18 | Irrigation Wells | IW 2 | 14.8°c | | | | | | H7 |
| 5 | | 7:25 | Irrigation Wells | IW 3 | 14.6°c | | | | | | H7 |
| 6 | | 7:36 | Irrigation Wells | IW 4 | 14.2°c | | | | | | H7 |
| 7 | | 7:45 | Irrigation Wells | IW 5 | 15.0°c | | | | | | H7 |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

**CHAIN OF CUSTODY RECORDING**

| | Signature | Company | Received Date & Time | Relinquished Date & Time | Logged in By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| | | F&R AG SERVICES | [redacted] | | | |
| 2nd | | DENELE ANALYTICAL, INC. | 12-19-23 10:30 | [redacted] | | T353003 |

*(LABORATORY USE ONLY)*

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

# Compliance Analysis Report

**Customer**
Wickstrom Dairy/Valsigna Farms
P.O. Box 880
Hilmar, CA 95324

**Grower**

**Consultant:**
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.0c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| IW 5 | T3353003G | Electrical Conductivity (EC) | EPA 120.1 and | 1.00 | mmhos/cm |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 38.8 | mg/L |
| | | Total Dissolved Solids (TDS) | SM 2540 C-1997 and | 605 | mg/L |
| | | Total Nitrogen (TN) | SM 4500-NH3 D,C | 39.0 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 643 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | Sample mg/L | Dup mg/L | RPD % |
|---|---|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | | | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 1.07 (T3352006A) | 1.06 (Dup) | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 109 (T3353012A) | 109 (T3353012A) | | | 0.055 |
| Total Dissolved Solids (TDS) | 105 | | | 200 (T3352010A) | 205 (Dup) | 2.47 |
| Total Nitrogen (TN) | 102 | | | 0.280 (T3353003G) | 0.280 (null) | 0.000 |

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P.O. Box 880
Hilmar, CA 95324

**Grower**

**Consultant:**
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.2c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| IW 4 | T3353003F | Electrical Conductivity (EC) | EPA 120.1 and | 1.02 | mmhos/cm |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 38.3 | mg/L |
| | | Total Dissolved Solids (TDS) | SM 2540 C-1997 and | 615 | mg/L |
| | | Total Nitrogen (TN) | SM 4500-NH3 D,C | 38.3 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 651 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | Sample mg/L | Dup mg/L | RPD % |
|---|---|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | | | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 1.07 (T3352006A) | 1.06 (Dup) | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 109 (T3353012A) | 109 (T3353012A) | | | 0.055 |
| Total Dissolved Solids (TDS) | 105 | | | 200 (T3352010A) | 205 (Dup) | 2.47 |
| Total Nitrogen (TN) | 99.1 | | | 0.000 (T3345031B) | 0.000 (null) | |

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**Grower**

**Consultant:**
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.6c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| IW 3 | T3353003E | Electrical Conductivity (EC) | EPA 120.1 and | 1.01 | mmhos/cm |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 37.8 | mg/L |
| | | Total Dissolved Solids (TDS) | SM 2540 C-1997 and | 595 | mg/L |
| | | Total Nitrogen (TN) | SM 4500-NH3 D,C | 37.8 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 648 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | Sample mg/L | Dup mg/L | RPD % |
|---|---|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | | | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 1.07 (T3352006A) | 1.06 (Dup) | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 109 (T3353012A) | 109 (T3353012A) | | | 0.055 |
| Total Dissolved Solids (TDS) | 105 | | | 200 (T3352010A) | 205 (Dup) | 2.47 |
| Total Nitrogen (TN) | 99.1 | | | 0.000 (T3345031B) | 0.000 (null) | |

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**Grower**

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.8c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| IW 2 | T3353003D | Electrical Conductivity (EC) | EPA 120.1 and | 1.00 | mmhos/cm |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 38.6 | mg/L |
| | | Total Dissolved Solids (TDS) | SM 2540 C-1997 and | 570 | mg/L |
| | | Total Nitrogen (TN) | SM 4500-NH3 D,C | 41.4 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 641 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | Sample mg/L | Dup mg/L | RPD % |
|---|---|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | | | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 1.07 (T3352006A) | 1.06 (Dup) | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 109 (T3353012A) | 109 (T3353012A) | | | 0.055 |
| Total Dissolved Solids (TDS) | 105 | | | 200 (T3352010A) | 205 (Dup) | 2.47 |
| Total Nitrogen (TN) | 99.1 | | | 0.000 (T3345031B) | 0.000 (null) | |

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**Grower**

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.8c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| IW 1 | T3353003C | Electrical Conductivity (EC) | EPA 120.1 and | 1.01 | mmhos/cm |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 37.8 | mg/L |
| | | Total Dissolved Solids (TDS) | SM 2540 C-1997 and | 590 | mg/L |
| | | Total Nitrogen (TN) | SM 4500-NH3 D,C | 37.8 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 648 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | Sample mg/L | Dup mg/L | RPD % |
|---|---|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | | | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 1.07 (T3352006A) | 1.06 (Dup) | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 109 (T3353012A) | 109 (T3353012A) | | | 0.055 |
| Total Dissolved Solids (TDS) | 105 | | | 200 (T3352010A) | 205 (Dup) | 2.47 |
| Total Nitrogen (TN) | 99.1 | | | 0.000 (T3345031B) | 0.000 (null) | |

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PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| DW 7 | T3353003B | Electrical Conductivity (EC) | EPA 120.1 and | 1.01 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 35.9 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 648 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 106 (T3353008A) | 100 (T3353008A) | 5.20 |

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ELAP Certificate No.: 2714

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**Grower**

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Lourenco
RECEIVED DATE: 12/19/23 10:30 AM
SUBMITTED BY: KK
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.8c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| DW 6 | T3353003A | Electrical Conductivity (EC) | EPA 120.1 and | 1.01 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 36.0 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 644 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.2 | 99.4 (T3352006A) | 98.2 (T3352006A) | 1.21 |
| Electrical Conductivity (EC) | 98.9 | | | 0.750 |
| Nitrate Nitrogen (NO3-N) | 105 | 106 (T3353008A) | 100 (T3353008A) | 5.20 |

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